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as it requires a unique skillset. You need to have proper architecture to build software, a design and the necessary components that will allow you to get started with the coding. You then have to validate and commercialise everything before it goes into the hands of developers and, ultimately, the consumers themselves. It all boils down to sufficient investment. But there is a limit to which everyone can invest in software and often it is only for a particular period of time.”

“Open source is no free lunch as somebody is indeed paying for it. When we use the term open source, we mean that we are enabling a shared software development model where we want to collaborate with similar entities that want to invest in the development of software. Hardware is also a fundamental aspect of building software, as without proper hardware in place software cannot be developed. We need low cost hardware that can be afforded by everybody. This is why the Intel ecosystem picked up, because everyone used a personal computer and these systems did not require a huge investment. Over time, embedded systems have also evolved and reduced a lot of the burden of software development. With the entry of Android all the necessary components of open source development have been stitched together,” he says.



Biju K. Nair,
founding partner,
Legalitech

“Today, things have evolved tremendously. We are seeing multi-core architecture and multi-core programming even in a simple smart watch today. It requires a great amount of skill to do it all. Not only are you not going to work for free while working on open source software, but your contribution to open source will also provide you with the relevant validation and introduce you to new features. Hence, companies always try to build an ecosystem for open source development as it tends to be more robust and reliable. It is time to define the hardware ecosystem that we are going to enter into, and the kind of software skillsets that are required, in order to assess what we need to build the community around. We also need to find whether it can scale and sustain on its own without a huge sum of money being pumped from big organisations,” adds Khasim.

The legal angle

Biju K. Nair, founding partner, Legalitech, says: “There is also a very important legal aspect to open source from an intellectual property (IP) perspective. IP is different in the Metaverse from how it is considered in the traditional ecosystem. So this creates opportunities to maintain and safeguard the IP within the Metaverse. In the last two years the whole globe

has woken up to the vulnerability issues. Everyone today, including the US government, is trying to spend more money on the detection of vulnerabilities and securing them. Working together as a community can allow us to solve these issues together.”

The advantages of open source in the Metaverse

So what is the main foundation and strength of open source, which will play a key role in the development of the Metaverse?

“Open source is both flexible and agile. It is important to note that the advantages of using open source far outweigh the disadvantages. GitHub is the largest open source community in the world or, you can say, it is the home of open source itself. At GitHub our focus is to continuously improve the platform and add features so that the collaboration and open source projects can thrive,” says Gandhali Samant.

“Today, every new security feature that we build is available for free for public repositories. It is the main source of developing open source software. If you look up the top 10 Metaverse projects on Google, you will find that they are all available on GitHub. The role of GitHub in the Metaverse will be very similar to the role that GitHub has played in the large scale success and adoption of technologies like TensorFlow for AI, or Docker and Kubernetes for containerisation, orchestration, Hyperledger, etc. GitHub will continue to provide the best platform for software development,” she adds.

But are there any limitations and concerns for the collaborators?

According to Samant, “On one side we have multiple open source projects, while on the other we have tech giants like Google and Microsoft that are already investing big dollars in open source. No one company holds the monopoly for any component today. They are bound to come together and work, as each one